

Welcome

To

Dhaka International University
DEPARTMENT OF CSE

Instructor : **ANTU CHOWDHURY**

Batch : **93(1st Semester)**

Course Code : **0613-101**

Course Title : **Structured Programming Languages**

Group: 01

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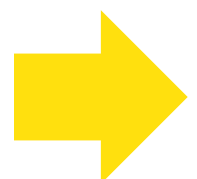


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Slide Title:



- **Title: Matrix Addition & Subtraction in C Programming**

Introduction

Definition of 2D Matrices:

- A 2D matrix is a collection of numbers arranged in rows and columns.

Rows and Columns:

- Rows: Horizontal arrangement of elements in a matrix.
- Columns: Vertical arrangement of elements in a matrix.

Example

1	2	3
4	5	6
7	8	9

2d matrix

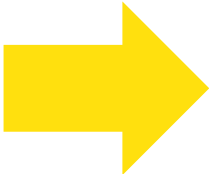
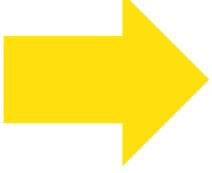
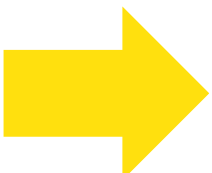
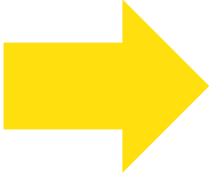
```
int matrix[3][3] =  
{  
    {1, 2, 3},  
    {4, 5, 6},  
    {7, 8, 9}  
};
```

2d array

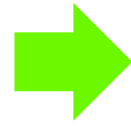
Problem name

Matrix Addition & Subtraction

To Do

-  **We need two 2D array as matrix.**
-  **Subtract corresponding elements of two matrices.**
-  **Add corresponding elements of two matrices.**
-  **Output the Addition & Subtraction result.**

ALGORITHM



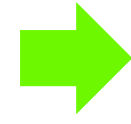
Input:

Prompt the user to enter the number of rows and columns for matrices A and B.



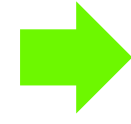
Input Elements for Matrix A:

Use nested loops to accept elements for matrix A from the user.



Input Elements for Matrix B:

Use nested loops to accept elements for matrix B from the user.



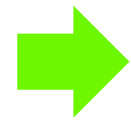
Print Matrix A:

Display the elements of matrix A.



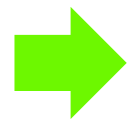
Print Matrix B:

Display the elements of matrix B.



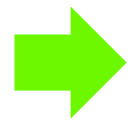
Matrix Addition:

Use nested loops to add corresponding elements of matrices A and B, storing the result in matrix C.



Print Matrix C (Result of Addition):

Display the elements of matrix C, which represents the result of matrix addition.



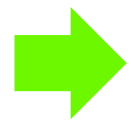
Matrix Subtraction:

Use nested loops to subtract corresponding elements of matrix B from A, storing the result in matrix C.



Print Matrix C (Result of Subtraction):

Display the elements of matrix C, which represents the result of matrix subtraction.



End:

End the program.

Code Structure:

```
#include<stdio.h>
int main ()
{
    int i,j,noofrows,noofcols;
    int A[10][10],B[10][10],C[10][10] ;
    printf("Enter the no of rows and cols:");
    scanf("%d %d",&noofrows,&noofcols);
    printf("Enter elements for be A matrix.\n");
    for(i=0; i<noofrows; i++)
        //scanner
    {
        for (j=0; j<noofcols; j++)
        {
            printf("A[%d][%d]:",i,j);
            scanf("%d",&A[i][j]);
        }
        printf("\n");
    }
    printf("Enter elements for be B matrix.\n");
    for(i=0; i<noofrows; i++)
        //scanner
    {
        for (j=0; j<noofcols; j++)
        {
            printf("B[%d][%d]:",i,j);
            scanf("%d",&B[i][j]);
        }
        printf("\n");
    }
    //print of A matrix
    printf("A: ");
    for (i=0; i<noofrows; i++)
```

```
    for (i=0; i<noofrows; i++)
    {
        printf("\t");
        for(j=0; j<noofcols; j++)
        {
            printf("%d ",A[i][j]);
        }
        printf("\n");
    }
    //print of B matrix
    printf("B:");
    for (i=0; i<noofrows; i++)
    {
        printf("\t");
        for(j=0; j<noofcols; j++)
        {
            printf("%d ",B[i][j]);
        }
        printf("\n");
    }
    //addition of matrix
    for (i=0; i<noofrows; i++)
    {
        for(j=0; j<noofcols; j++)
        {
            C[i][j] =A[i][j] +B[i][j] ;
        }
    }
    //print in C matrix
    printf("\n");
    printf("A+B:");
```

```
    printf("\n");
    printf("A+B:");
    for (i=0; i<noofrows; i++)
    {
        printf("\t");
        for(j=0; j<noofcols; j++)
        {
            printf("%d ",C[i][j]);
        }
        printf("\n");
    }
    for (i=0; i<noofrows; i++)
    {
        for(j=0; j<noofcols; j++)
        {
            C[i][j] =A[i][j] - B[i][j] ;
        }
    }
    printf("\n");
    printf("A+B:");
    for (i=0; i<noofrows; i++)
    {
        printf("\t");
        for(j=0; j<noofcols; j++)
        {
            printf("%d ",C[i][j]);
        }
        printf("\n");
    }
}
```

Explanation

Code Walkthrough

```
int i,j,noofrows,noofcols;  
int A[10][10],B[10][10],C[10][10] ;
```

Declaring the variables

```
printf("Enter elements for be A matrix.\n")  
for(i=0; i<noofrows; i++)  
    //scanner  
{  
    for (j=0; j<noofcols; j++)  
    {  
        printf("A[%d][%d]:",i,j);  
        scanf("%d",&A[i][j]);  
    }  
    printf("\n");  
}
```

```
Enter elements for be A matrix.  
A[0][0]:1  
A[0][1]:2  
  
A[1][0]:3  
A[1][1]:4
```

```
printf("Enter elements for be B matrix.\n")  
for(i=0; i<noofrows; i++)  
    //scanner  
{  
    for (j=0; j<noofcols; j++)  
    {  
        printf("B[%d][%d]:",i,j);  
        scanf("%d",&B[i][j]);  
    }  
    printf("\n");  
}
```

```
Enter elements for be B matrix.  
B[0][0]:5  
B[0][1]:6  
  
B[1][0]:7  
B[1][1]:8
```

Explanation

```
//print of A matrix  
printf("A: ");  
for (i=0; i<noofrows; i++)  
{  
    printf("\t");  
    for(j=0; j<noofcols; j++)  
    {  
        printf("%d ",A[i][j]);  
    }  
    printf("\n");  
}
```

```
//print of B matrix  
printf("B:");  
for (i=0; i<noofrows; i++)  
{  
    printf("\t");  
    for(j=0; j<noofcols; j++)  
    {  
        printf("%d ",B[i][j]);  
    }  
    printf("\n");  
}
```

Code Walkthrough

A: 1 2
 3 4

B: 5 6
 7 8

Code Walkthrough

```
//addition of matrix
for (i=0; i<noofrows; i++)
{
    for(j=0; j<noofcols; j++)
    {
        C[i][j] =A[i][j] +B[i][j] ;
    }
}
//print in C matrix
printf("\n");
printf("A+B:");
for (i=0; i<noofrows; i++)
{
    printf("\t");
    for(j=0; j<noofcols; j++)
    {
        printf("%d ",C[i][j]);
    }
    printf("\n");
}
```



```
A+B:      6  8
          10 12
```

```
for (i=0; i<noofrows; i++)
{
    for(j=0; j<noofcols; j++)
    {
        C[i][j] =A[i][j] - B[i][j] ;
    }
}
//print in C matrix
printf("\n");
printf("A-B:");
for (i=0; i<noofrows; i++)
{
    printf("\t");
    for(j=0; j<noofcols; j++)
    {
        printf("%d ",C[i][j]);
    }
    printf("\n");
}
```



```
A-B:      -4 -4
          -4 -4
```

CODE OUTPUT:

```
Enter elements for be B matrix.
```

```
B[0][0]:5
```

```
B[0][1]:6
```

```
B[1][0]:7
```

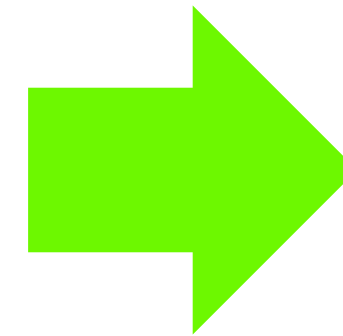
```
B[1][1]:8
```

```
A:      1  2  
        3  4
```

```
B:      5  6  
        7  8
```

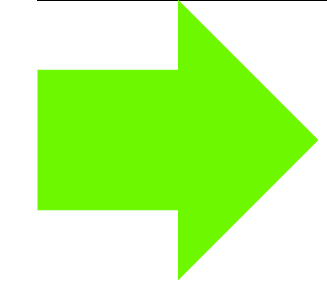
```
A+B:     6  8  
       10 12
```

```
A-B:    -4 -4  
       -4 -4
```

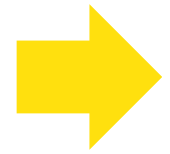


OUTPUT

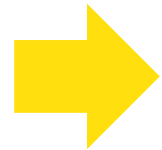
Conclusion



• Summary:



- We successfully demonstrated the addition and subtraction operations on matrices using C programming.



- Understanding basic concepts of matrices and programming structures is crucial for solving such problems effectively.

